FOUNDATIONS OF GENERATIVE AI (ICT4403)

Assignment 3: Neural Machine Translation Across Transformer Architectures

Submission Cover Sheet

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Assignment Title	ICT4403: Foundations of Generative AI — Assignment 3
Title of Developed Autoregressive System	Comparative English-to-French Neural Machine Translation & Ethical Auditing System (Basic Transformer, mT5, LLaMA, Gemma)
Selected Language Pair	Source: English (en) --> Target: French (fr)
Name of Dataset Used	OPUS Books Parallel Corpus (Helsinki-NLP/opus_books, en-fr)
Dataset Source Link	https://huggingface.co/datasets/Helsinki-NLP/opus_books (OPUS Portal: https://opus.nlpl.eu/Books.php)
GitHub Repository Link	https://github.com/Lemonseeds/gen-ai-proj-2
Primary Deliverables in Repository	• FGAI_Assignment3.ipynb (Fully executed master notebook with inline outputs & plots) • main.py & modular Python package in src/ (data, models, train_and_eval) • Saved PyTorch model checkpoint in models/basic_transformer.pt • Numerical metrics (CSV/JSON), test set translation samples, and 5 analytical figures in results/ • Comprehensive README.md with mathematical formulations, evaluation tables, and ethical audit • Submission PDF & Word reports: AyrtonJosephPulikottil_230953448_FGAI_Assignment3.pdf

Executive Summary of Empirical Findings & Architectural Comparison

This project implements an end-to-end comparative Neural Machine Translation (NMT) and ethical auditing system evaluating four distinct Transformer models across encoder–decoder and decoder-only paradigms on the publicly available OPUS Books parallel literature corpus (en -> fr):

1. Basic Transformer (Encoder–Decoder, Trained from Scratch): Custom 6-layer PyTorch Seq2Seq model (d=192, H=6) trained for 8 epochs on CPU. Achieved test BLEU = 0.91, chrF = 17.34, with the lowest per-sentence inference latency (37.6 ms/sentence) and smallest memory footprint (2.54M parameters).

2. mT5-Small (google/mt5-small, Encoder–Decoder, Pretrained & Fine-Tuned): Multilingual T5 (300.18M parameters) adapted via French/English sub-vocabulary projection (slicing 250,112 SentencePiece tokens down to 5,587 active subwords) and seq2seq fine-tuning. Achieved test BLEU = 1.16, chrF = 13.03, latency = 82.1 ms/sentence.

3. LLaMA-135M (LlamaForCausalLM / SmolLM2-135M, Decoder-Only): Adapted via 3-Shot In-Context Prompting and top-layer LoRA fine-tuning (r=8, alpha=16). Achieved BLEU = 0.50, chrF = 14.25 (3-shot prompting) and chrF = 14.32 (LoRA fine-tuned), latency = 598.7 ms/sentence.

4. Gemma-3-270M (Gemma3ForCausalLM / gemma-3-270m-it, Decoder-Only): Adapted via 3-Shot In-Context Prompting and LoRA fine-tuning. Achieved the highest translation fidelity across all benchmarks: BLEU = 12.24, chrF = 32.84, latency = 625.2 ms/sentence.

Sampling-Temperature Dynamics:
Testing across T in {0.0, 0.3, 0.7, 1.2} demonstrated that deterministic greedy decoding (T=0.0) maximizes lexical and morphological fidelity for NMT (Gemma chrF=36.4 at T=0.0 vs. 24.1 at T=1.2), while elevated temperature (T >= 0.7) increases lexical diversity (Distinct-2 > 0.90) at the expense of semantic drift and syntactic hallucination.

Responsible & Ethical AI Auditing:
Diagnostic probing across 8 targeted linguistic pairs revealed systemic occupational gender stereotyping (defaulting 'doctor'/'engineer' to masculine and 'nurse'/'secretary' to feminine), pronominal bias ('they' -> 'ils'), and risk of safety-critical negation inversion in medical prompts. Counterfactual data augmentation, formality-controlled prompting, and constrained decoding are proposed and analyzed as mitigation strategies.